

Lab 3: Architectural Justification

System: Hostel Maintenance & Issue Ticketing System

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Problem Statement

PROBLEM STATEMENT #07 | Campus & Academic Operations — Hostel Maintenance & Issue Ticketing System

Architecture Selection

We chose Layered Architecture for the Hostel Maintenance & Issue Ticketing System.

Architectural Choice

The system is organized into three layers: a Presentation Layer (Resident & Warden UI), a Business Layer (Ticket Workflow Manager, Assignment Engine, Notification & Escalation Service, Auth & Access Control), and a Data Layer (Ticket Database). All requests flow top-down through well-defined interfaces; the Presentation Layer never accesses the Data Layer directly.

Two Reasons for This Choice

1. Enforceable ticket lifecycle (FR-003): The ticket state machine (Logged → Assigned → In Progress → Resolved → Closed) must be advanced only by the warden or assigned staff, with invalid jumps blocked. A layered design lets the Business Layer's Ticket Workflow Manager act as the single authority over these transitions, so no client or database write can skip a stage or fake an SLA record.

2. Right-sized for a single-institution workload: This is one hostel/campus with steady, predictable ticket volume rather than independently scaling domains. Layered architecture gives clean separation of concerns without the operational cost (service discovery, distributed transactions, network latency) that Microservices would add for this scale, and without concentrating all traffic on one tier the way a plain Client-Server split would.

Security Advantage

Role-based access (NFR-002 — residents see only their own tickets, wardens only their own block) is enforced by routing every Business Layer call through one Auth & Access Control component via the AccessCheck interface. This gives a single choke point for authorization instead of duplicating access checks across every UI screen, where a check could be missed or bypassed.

Performance Benefit

Because the Presentation Layer never queries the Data Layer directly, the Ticket Workflow Manager can push incremental status updates to clients from one place instead of every client polling the database. This keeps FR-005's 5-second status-sync target achievable even at peak load, such as when many residents log tickets at the start of a semester.

System Components & Interface Specifications

Components List

Sr. No	Component	Layer	Maps to Requirements
1	Resident & Warden UI	Presentation	FR-001, FR-004, FR-005 (log tickets, confirm resolution, view live status)
2	Ticket Workflow Manager	Business	FR-003, FR-005 (lifecycle state machine, status push to clients)

Sr. No	Component	Layer	Maps to Requirements
3	Assignment Engine	Business	FR-002 (auto-assign new ticket to free, category-matched staff)
4	Notification & Escalation Service	Business	NFR-001 (SMS/email escalation for stalled high-priority tickets)
5	Auth & Access Control	Business (Crosscutting)	NFR-002 (role-based access for residents and wardens)
6	Ticket Database	Data	Persistence for tickets, attachments, and status history

Interfaces

#	Interface	Provided by	Required by	Purpose
1	TicketSubmission	Ticket Workflow Manager	Resident & Warden UI	Resident submits a new ticket with room tag, photo, and severity
2	StatusQuery	Ticket Workflow Manager	Resident & Warden UI	Read-only live status and history lookups for residents and wardens
3	AssignTicket	Assignment Engine	Ticket Workflow Manager	Route a newly logged ticket to a free, category-matched staff member
4	AccessCheck	Auth & Access Control	Ticket Workflow Manager, Assignment Engine, Notification & Escalation Service	Enforce role-based access before any business-layer operation
5	EscalationTrigger	Notification & Escalation Service	Ticket Workflow Manager	Fire SMS/email alert when a high-priority ticket exceeds the 24-hr SLA
6	TicketDataAccess	Ticket Database	Ticket Workflow Manager, Assignment Engine	Persist and retrieve ticket records, attachments, and status history

Component Diagram

